Statistical Thinking for Data Science

Week 1: Introduction to Data Science

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Lecture notes for students of Statistical Thinking for Data Science.

Week 1: Introduction to Data Science

Learning Objectives

By the end of this week you should be able to:

- say what data science is, in a sentence that does not use the word *data* twice;
- explain why so much data exists now when it did not twenty years ago;
- sort a question into one data science can answer, one it cannot, and one it should not be asked to answer;
- describe what statistical thinking contributes that computing alone does not;
- state the responsibilities that come with holding data about people.

1.1 What data science is

A mining engineer has twenty years of blast records. A hospital in Tarkwa has ten years of admissions. A telecommunications company has every call its network carried last month. In all three cases somebody is sitting on a quantity of data that nobody has the time to read, and somewhere inside it is the answer to a question that matters.

Data science is the work of getting that answer out, and of knowing how much to trust it when it arrives.

Definition 1.1 (Data science)**.** The practice of answering questions with data: deciding what to measure, collecting it, cleaning it, describing it, modelling it, and communicating what it does and does not support.

That definition is deliberately plain. You will meet more fashionable ones, usually involving a diagram of three overlapping circles. The three circles are not wrong: the work does sit between **mathematics and statistics**, **computing**, and **the subject the data is about**. What the diagram hides is that the third circle is the one most often missing. A model of mill throughput built by somebody who has never stood next to a mill will fit the numbers and miss the point.

1.1.1 What data science is not

- **It is not simply a large amount of data.** Size is a practical problem, not an intellectual one. A badly collected million rows is worth less than a well collected hundred.
- **It is not only machine learning.** Prediction is one task among several. Describing what happened, estimating a quantity, and testing a claim are others, and they are more often what an employer actually needs.
- **It is not a substitute for knowing your subject.** The data will not tell you that a sensor was offline for three weeks in March. A technician will.
- **It is not neutral.** Someone chose what to measure, who to measure it on, and what to leave out. Those choices are in the results, whether or not anyone mentions them.

Data science is a team activity more often than a solo one. The person who understands the process, the person who can manage the data, and the person who understands uncertainty are rarely the same person. Your value is partly in being able to talk to the other two.

1.2 Why there is suddenly so much data

Data used to be expensive. Someone had to decide a measurement was worth taking, assign a person to take it, and pay for the paper it was written on. That cost acted as a filter: data existed because somebody wanted it.

Three things removed the filter.

1. **Measurement became automatic.** Equipment now records itself. A haul truck logs load, fuel, engine temperature and position every few seconds, whether anyone reads it or not.
2. **Everybody carries a sensor.** A phone is a location logger, a camera and a payment terminal. Mobile money alone generates a record of a large share of the country's small transactions, which thirty years ago left no trace at all.
3. **Storage became nearly free.** When keeping data costs almost nothing, the default changes from *discard unless needed* to *keep in case*.

This matters for a reason that is easy to miss. Most of the data you will work with was **not collected to answer your question**. It is a by-product of something else: billing, compliance, maintenance, or a government return. Data collected as a by-product is still useful, but it carries the shape of the process that produced it.

> **A habit to start now.** Before analysing any dataset, write down in one sentence why it was collected and by whom. If you cannot, you do not yet know enough to analyse it.

1.3 The questions data can answer

It helps to sort questions by what kind of answer they demand.

Kind	Example in your field
Descriptive	How much did the plant produce each month last year?
Diagnostic	Why was output lower in the rainy season?
Predictive	How much will it produce next month?
Causal	Would changing the grinding circuit raise recovery, or is the association with recovery caused by ore grade?
Prescriptive	Given cost, risk and capacity, what should we actually do?

They are ordered by difficulty, and the jump from predictive to causal is the big one. A model can predict perfectly and tell you nothing about what to change: umbrellas predict rain, but confiscating umbrellas will not keep you dry.

1.3.1 Questions data cannot answer

Three kinds, worth recognising early.

- **Questions with no data behind them.** Nobody recorded the thing you need. The honest answer is to say so, then to design a way of collecting it.
- **Questions where the data cannot separate the explanations.** If every plant that adopted the new reagent also upgraded its pumps the same year, no amount of analysis will tell you which change did the work.
- **Questions that are not about facts.** How safe is safe enough? Whose time is worth more? Data informs those decisions; it does not make them. Being asked for a number that settles a value judgement is common, and declining politely is part of the job.

1.4 Where statistical thinking comes in

Suppose you compute the mean grade of ore from forty samples and get $\bar{x} = 2.41$ g/t. The number is exact. What it tells you about the deposit is not. Another forty samples would have given a different number. Statistical thinking is the discipline of carrying that second sentence around with you.

It amounts to four habits:

1. **Expect variation.** Two measurements of the same thing differ. The question is never whether numbers differ but whether they differ by more than they would anyway.
2. **Ask where the data came from.** Who is in the dataset, who is missing, and would the missing ones have answered differently?
3. **Quantify uncertainty.** An estimate without a range is half an answer.
4. **Distrust a pattern you went looking for.** Examine enough variables and something will look significant by luck alone. We return to this in Week 10.

Example 1: why a ninety-five percent accurate test is not ninety-five percent right

An automated system inspects castings. Historically 2% of castings are faulty. The system detects 90% of faulty castings, and wrongly flags 5% of sound ones. A casting has just been flagged. What is the probability that it is genuinely faulty?

Solution

Take 1000 castings. Of these, 20 are faulty and 980 are sound.

- Faulty and flagged: $20 \times 0.90 = 18$.
- Sound but flagged: $980 \times 0.05 = 49$.

So 67 castings are flagged, of which 18 are truly faulty:

$$P(\text{faulty} \mid \text{flagged}) = \frac{18}{18 + 49} = \frac{18}{67} \approx 0.27.$$

Fewer than three flagged castings in ten are actually faulty, even though the system is right about 95% of the time overall. The reason is that sound castings are so much more numerous that their few false alarms swamp the true ones.

This single calculation, which we will meet again properly in Week 8, is the clearest example of why intuition needs statistics. Nobody guesses 27%.

Example 2: a sample that answers a different question

A district wants the average time households spend fetching water. A survey is conducted by telephone. The average among those reached is 1.8 hours per day.

Suppose 70% of households have a phone, and that households without a phone are poorer and, in truth, spend 3.2 hours. The true district average is

$$0.70 \times 1.8 + 0.30 \times 3.2 = 1.26 + 0.96 = 2.22 \text{ hours},$$

so the survey understates the burden by 0.42 hours, about 19%, and understates it exactly for the households the intervention is meant to help.

Lesson

No amount of extra phone interviews fixes this. A bigger biased sample is simply a more confident wrong answer. Sampling is the subject of Week 3.

1.5 Seeing the data

Visualisation does three separate jobs, and confusing them produces bad graphics.

1. **Exploration**, for you. Quick, ugly, many of them. This is where you notice the impossible value, the gap where a sensor failed, the two clusters that turn out to be two shifts.
2. **Diagnosis**, for the model. Residual plots and the like, which tell you whether what you fitted is defensible.
3. **Communication**, for somebody else. One careful figure that makes a single comparison unmissable.

A figure made for the first job and shown for the third is the commonest fault in student reports, and in consultancy reports too. We spend Week 6 on this.

1.6 What the work demands of you

You will hear long lists of required skills. The short version is four things:

- **Statistics**, so you know what the numbers support.
- **Computing**, enough to get data into a usable state and to automate what you repeat. In this course we use R or Python; the ideas do not depend on which.
- **The subject matter**, or close access to someone who has it.
- **Communication.** An analysis nobody acts on has failed, however elegant.

To that I would add one more, which is less often listed: **the willingness to say the data does not support the conclusion somebody wants**. That is the whole value of the role. Anybody can produce a number.

> If you are not going to be a data scientist, you still need enough of this to be a critical consumer: to ask where the data came from, what was left out, and how sure anyone really is. Most of the damage done with data is done to people who could not ask those three questions.

1.7 Data about people

Much of what you will analyse concerns people: patients, customers, students. Three obligations follow, and they are not optional extras.

- **Consent and purpose.** Data gathered for one purpose should not quietly be used for another. In Ghana this is law, not etiquette: the Data Protection Act, 2012 (Act 843) governs how personal data may be collected and used.
- **Anonymity is harder than removing names.** A programme, an age and a hall of residence can identify one student even with no name attached. This is why the class survey you will meet shortly asks for age in bands.
- **Harm can be statistical.** A model that is accurate on average can be systematically wrong for a group that is rare in the training data, and the people affected are rarely the ones who built it.

Our own dataset. During this course you will be invited to complete an anonymous wellbeing questionnaire, and the class will analyse the results together. It is voluntary and carries no marks. We use it because analysing data you collected yourself teaches something that a clean textbook dataset cannot: you will know exactly who is in it, who is missing, and what the numbers can honestly be asked.

1.8 Summary

Week 1 in five lines

- Data science answers questions with data, and says how much to trust the answer.
- Most data you meet was collected for some other purpose, and carries its shape.
- Sort the question first: descriptive, predictive and causal are different jobs.
- Variation is the subject, not the nuisance.
- Data about people brings legal and ethical duties that start before the analysis.

Exercises

1. Define data science in one sentence, in your own words, without using the word *data* more than once.

2. Name a dataset that exists at UMaT because of some administrative process rather than because anyone intended to study it. What question could it answer, and what would it systematically miss?

3. Classify each as descriptive, predictive, causal or not answerable with data:
   (a) How many students sat PS 361 last year?
   (b) Will the number rise next year?
   (c) Does attending tutorials raise the pass rate?
   (d) Should tutorials be compulsory?

4. In Example 1, suppose the fault rate rises from 2% to 10% while the detection and false alarm rates stay the same. Recompute $P(\text{faulty} \mid \text{flagged})$ and explain the direction of the change.

5. In Example 2, what would the true average have to be among households without a phone for the telephone survey to be unbiased?

6. A colleague reports that machines serviced by the night shift fail more often, and proposes retraining that shift. Give two explanations that have nothing to do with the shift's competence, and say what you would need in order to separate them.

7. Find a chart in a newspaper or on a company website. Write down the comparison it is making, and one way it makes that comparison harder than it needs to be.

8. A dataset of student records has had names and index numbers deleted, and keeps programme, year, sex and hall. Explain why some students could still be identified, and suggest one change that would reduce the risk.

Before next week

Read Chapter 1 of *OpenIntro Statistics* (free at `https://www.openintro.org/book/os/`). Come with one question you would like to answer with data this semester, written on paper. In Week 2 we turn those questions into ones that can actually be answered.